

Lecture 2 Handout: The Potential Method

Intuitively, a **potential function argument** says:

If I start with some amount of water in a leaky bucket, then soon water must stop leaking out.



Ingredients of the argument:

1. Define the “unit of time” e.g. one iteration of an algorithm
2. Define a measure of the “amount of water in the bucket.”
This is the **potential function S_i** $\leftarrow$ amount of water in bucket at time i
3. Provide a(n) _____ bound on **S_0**
4. Provide a(n) _____ bound on **S_i** values
5. Prove that the bucket “leaks quickly”. I.e. that **S_i** decreases by at least some amount per unit time.
6. Use this to _____

Analyzing the Flipping Game via a Potential Function

1. Unit of time = _____
2. Define the **potential function** $S_i =$ _____
3. Note that $S_0 \leq$ _____
4. S_i can never be _____
5. At every turn the value of S_i decreases by _____
6. This implies that the total number of turns is at most _____

Does this prove the algorithm is *efficient*?

Analyzing Euclid's Algorithm with a Potential Function (attempt 1)

1. Unit of time = _____
2. Define the **potential function** $S_i =$ _____
3. $S_0 =$ _____
4. $S_i \geq$ _____
5. Each call the value of S_i decreases by _____
6. This implies that the total number of calls is at most _____

Does this prove the algorithm is *efficient*?

Analyzing Euclid's Algorithm with a Potential Function (attempt 2)

1. Unit of time = _____
2. Define the **potential function** $S_i =$ _____
3. $S_0 =$ _____
4. $S_i \geq$ _____
5. Each call the value of S_i decreases by _____
6. This implies that the total number of calls is at most _____

Does this prove the algorithm is *efficient*?

Let's convince ourselves that potential functions
 $S_i = \underline{\hspace{1cm}}$ and $S_i = \underline{\hspace{1cm}}$ give $\exp(n)$ running time bounds
(for time unit = 1 recursive call)

$S_i = \underline{\hspace{1cm}}$: What is an example of x, y values (≥ 10) such that
 S_i decreases by only 1?

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Analyzing Euclid's Algorithm with a Potential Function (attempt 3)

1. Unit of time = _____
2. Define the **potential function** $S_i =$ _____
3. $S_0 =$ _____
4. $S_i \geq$ _____
5. Each call the value of S_i decreases by _____
6. This implies that the total number of calls is at most _____

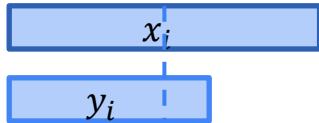
Does this prove the algorithm is *efficient*?

Analyzing Euclid's Algorithm via a Potential Function

Potential function: $S_i = \underline{\hspace{2cm}}$

Goal: $S_{i+1} \leq c S_i$

Case 1: $y_i > \underline{\hspace{1cm}}$

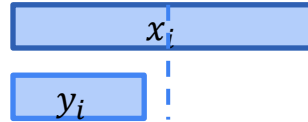


$S_{i+1} =$

Reminders about Euclid(x,y):

- $x > y \geq 0$
- $\text{Return}(\text{Euclid}(y, x \bmod y))$

Case 2: $y_i \leq \underline{\hspace{1cm}}$



$S_{i+1} =$

Either way, $S_{i+1} < \underline{\hspace{1cm}} S_i$

Analyzing Euclid's Algorithm via a Potential Function

Claim 2: If $s_{i+1} \leq \frac{1}{2} \cdot s_i$ for all i , then total # recursive calls is $O(n)$.

Proof: